Questions

Q1.

A student has a bar magnet, a piece of iron the same size as the magnet, and some paper clips.

Describe how the student could use these items to demonstrate temporary induced magnetism.

(3)

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(Total for question = 3 marks)

Q2.

A student investigates moments of forces.

Figure 14 shows the apparatus used.

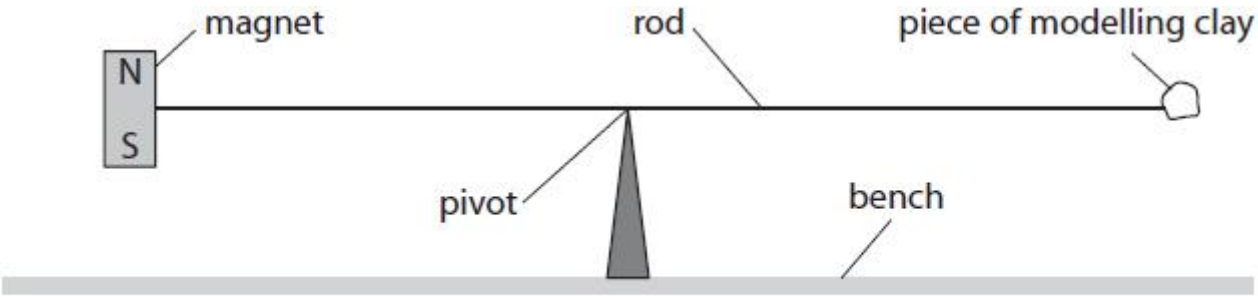

Figure 14

The pivot is under the centre of the rod.

A magnet is fixed to one end of the rod.

A piece of modelling clay is fixed to the other end of the rod.

The system is in equilibrium.

State the relationship between the moment of the weight of the magnet and the moment of the weight of the piece of modelling clay about the pivot.

(1)

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.....

(Total for question = 1 mark)

Q3.

Figure 10 shows a current-carrying wire between the poles of a magnet.

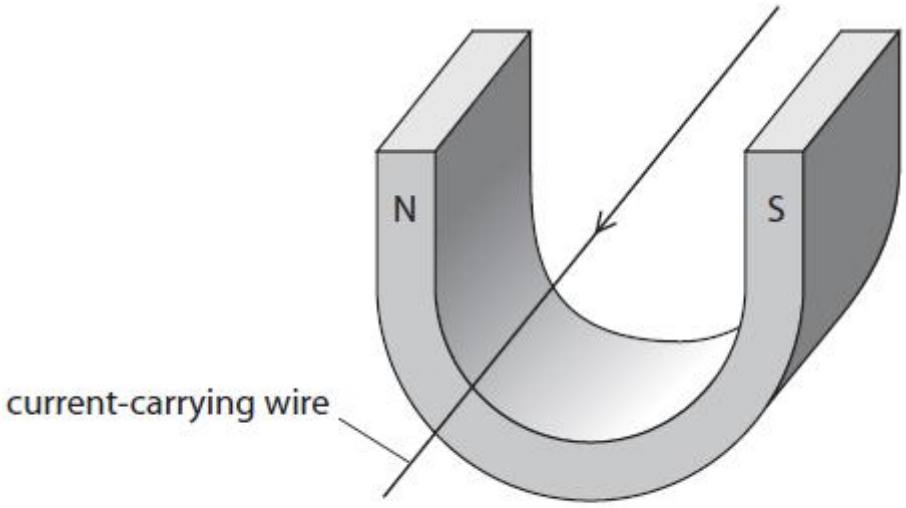


Figure 10

(i) The magnet and the wire each experience a force when there is a current in the wire.

(2)

1 State the direction of the force on the wire.

.....

2 State the direction of the force on the magnet.

.....

(ii) The force on the wire is 0.15 N.

The current in the wire is 2.7 A.
The magnet produces a field with a magnetic flux density of 0.50 T.
Calculate the length of the wire in the magnetic field.
Use an equation selected from the list of equations given at the end of the question paper.

(2)

length of the wire in the magnetic field = ... m

(Total for question = 4 marks)

Q4.

State how a uniform magnetic field may be obtained in a school laboratory.

(1)

.....

.....

(Total for question = 1 mark)

Q5.

Figure 14 shows a wire placed between the poles of a U-shaped magnet.

The wire is connected to a resistor and a battery.

The wire carries a current in the direction shown.

The wire is perpendicular to the magnetic field of the magnet.

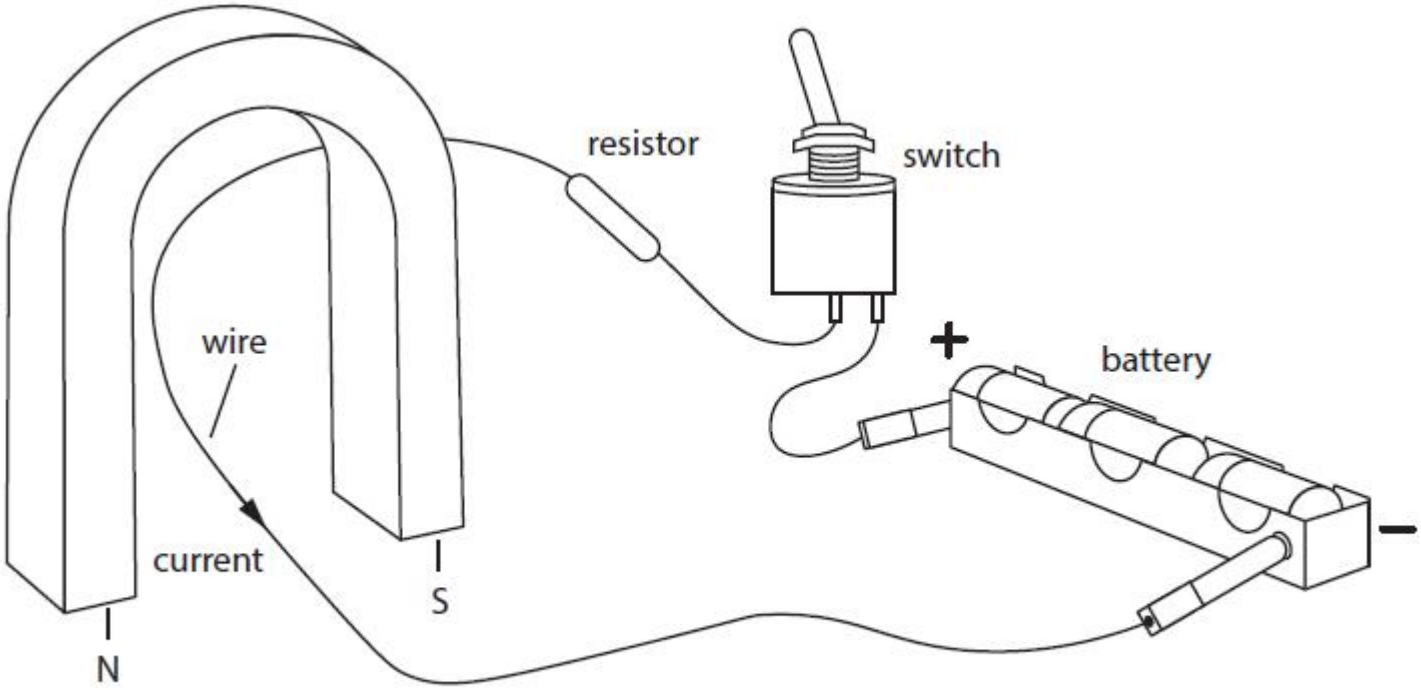


Figure 14

(i) Draw an arrow on Figure 14 to show the direction of the force, F, acting on the wire.

Label this arrow 'F'.

(1)

(ii) State **one** practical way of reversing the direction of force F.

(1)

.....

.....

(iii) In Figure 14

- current in the wire = 3.2 A
- length of wire in the magnetic field = 0.042 m
- magnitude of the force on the wire = 0.078 N

Calculate the magnitude of the magnetic flux density between the two poles of the magnet.

(2)

magnetic flux density = ... T

(Total for question = 4 marks)

Q6.

Figure 12 shows the directions of some plotting compass needles placed at different points near the Earth's surface.

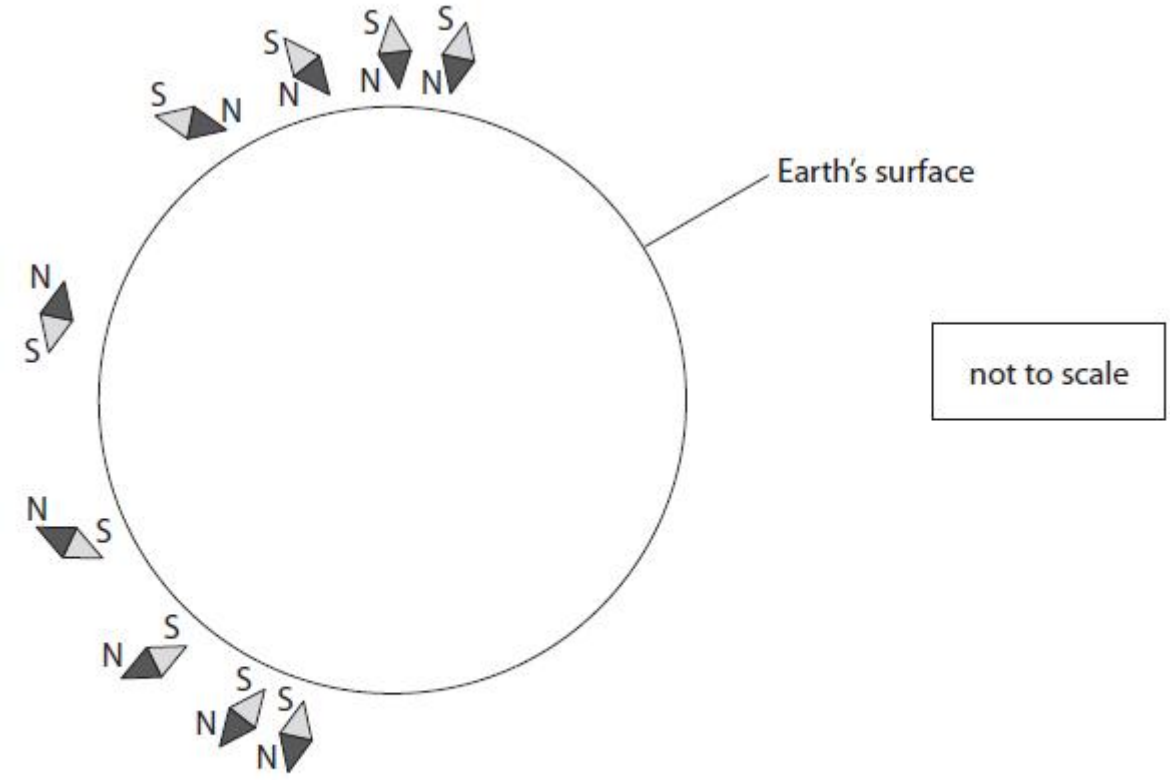


Figure 12

(i) Sketch, on Figure 12, the Earth's magnetic field outside and inside the Earth.

(2)

(ii) State which part of the Earth generates its magnetic field.

(1)

.....

.....

(Total for question = 3 marks)

Q7.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Figure 11 shows two nails, X and Y, hanging from a magnet.

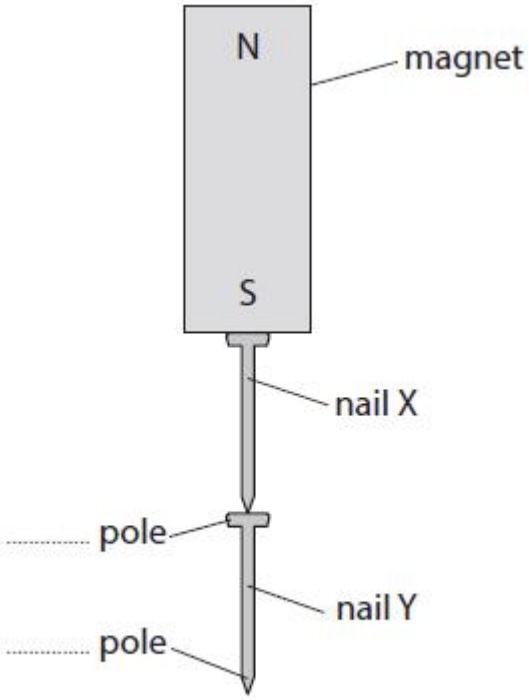


Figure 11

(i) Complete the labels in Figure 11 to show the N and S poles of nail Y.

(1)

(ii) Nail X is removed from the magnet and nail Y falls off nail X.

This shows that the magnet caused both nails to be

(1)

- ☒ **A** permanent magnets
- ☒ **B** temporarily charged
- ☒ **C** permanently charged
- ☒ **D** induced magnets

(Total for question = 2 marks)

Mark Scheme

Q1.

Question Number:	Answer	Additional Guidance	Mark
	a description to include:		**(3)** AO 1 2
	method of producing temporary induced magnetism (1)	place iron near / in contact with magnet / in magnetic field OR use magnet to pick up one paper clip OR use magnet to make iron a temporary magnet	
	method of demonstrating the magnetic properties of the temporary magnet (1)	paper clip(s) attracted to iron OR use first paper clip to pick up another paper clip	
	method of demonstrating magnetic effect is temporary (1)	remove magnet and paper clips no longer attracted / fall off OR wait some / short time and iron bar no longer picks up / attracts paper clips	

Q2.

Question number	Answer	Additional guidance	Mark
	(sum of) the clockwise moments = (sum of) the anticlockwise moments	moment of magnet = moment of modelling clay moments are equal (size)	**(1)** **AO1**

Q3.

Question number	Answer	Additional guidance	Mark
(i)	1 up(wards) (1) 2 down(wards) (1)	independent marks accept out(wards from the magnet) accept in(wards) / into (magnet) allow 1 mark for 1 down / in(wards) AND 2 up / out(wards)	**(2)** **AO1**

Question number	Answer	Additional guidance	Mark
(ii)	substitution (1) 0.15 = 0.5(0) x 2.7 x L(ength) rearrangement and evaluation (1) (length =) 0.11 (m)	alternative method re-arrangement (1) (length =) $\frac{F}{B \times I}$ Or (length =) $\frac{0.15}{0.5(0) \times 2.7}$ (substitution and) evaluation (1) (length =) 0.11 (m) allow any values that round to 0.11 e.g 0.111 accept 0.1 or 0.$\dot{1}$ (m) allow 1 mark for answer of 9 (with or without working) award full marks for correct answer without working.	**(2)** **AO2**

Q4.

Question number	Answer	Additional guidance	Mark
	(inside) a solenoid / long coil (with a current / power supply) (1)	give credit for diagrams accept: horseshoe magnet (between / using) pair of Magnadur / flat magnets (between / using) Helmholtz coils (between / using) two bar magnets, with unlike poles facing each other	**(1)** **AO1.2**

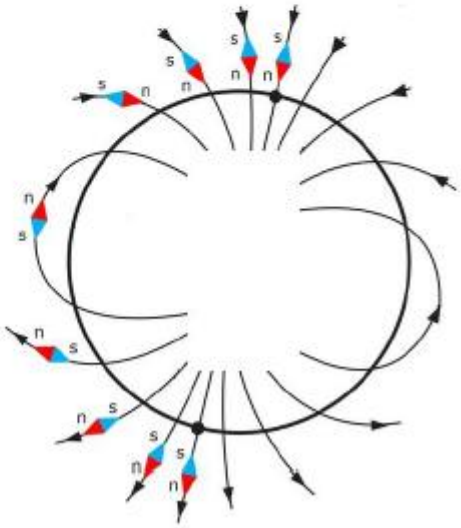
Q5.

Question number	Answer	Additional guidance	Mark
i	arrow pointing vertically up (1)	seen anywhere judge direction by eye	**(1)** **AO2.1**

Question number	Answer	Additional guidance	Mark
ii	statement (1) accept any clear **action** that will reverse the current **OR** accept any clear **action** that will reverse the poles	 swap the battery connections around turn the magnet around	**(1)** **AO2.2**

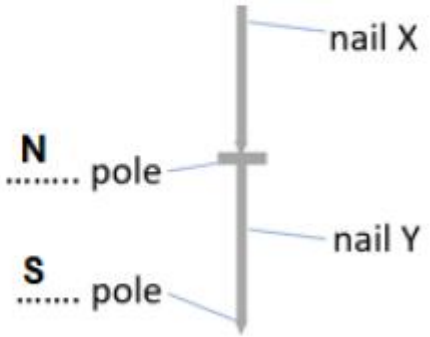
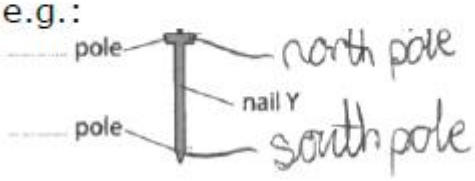
Question number	Answer	Additional guidance	Mark
iii	rearrangement and substitution (1) (B =) $\frac{0.078}{3.2 \times 0.042}$ evaluation (1) 0.58 (T)	(B =) $\frac{0.078}{0.1344}$ any number rounding to 0.6 (T) award full marks for the correct answer without working	**(2)** **AO2.1**

Q6.

Question number	Answer	Additional guidance	Mark
(i)	Sketch including any **two** from at least two field lines outside the Earth approximately aligning with compasses (1) at least two field lines continue inside the Earth towards imaginary poles (1) all arrows on lines drawn in the correct direction(s) outside the Earth (1) 	field lines need to have a gap inside the Earth ignore arrows on field lines inside the Earth	**2** **AO3.1**

Question number	Answer	Additional guidance	Mark
(ii)	(magnetic outer) **core** (1)	moving charges/ions	**(1)** **AO1.1**

Q7.

Question number	Answer	Additional guidance	Mark
(i)	both labels as shown (1)  	accept North / north for N South / south for S accept labels at the side e.g.:  	(1) AO1

Question number	Answer	Additional guidance	Mark
(ii)	**D** induced magnets B and C are incorrect because electric charge is not relevant in this context A is incorrect because permanent magnets would remain attracted		(1) AO1